L-2/T-2/CSE Date: 09/02/2025

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-2 B. Sc. Engineering Examinations 2023-2024

Sub: **CSE 213** (Software Engineering)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Question No. **1** is **Mandatory.**

Answer any **TWO** out of the remaining three questions.

1. Read the following case study carefully and then answer the questions given below.

A mental health clinic is using an information system named, Mentcare system, which is designed and used to maintain diagnostic and treatment related records for mental patient. Clinic staff log on to the system using a username and password. The system requires passwords to be at least eight letters long but allows any password to be set without further checking. A receptionist may transfer data from the Mentcare system to a national patient record system (PRS) database that is maintained by the health authority of the country. The information transferred may either be updated personal information (address, phone number, etc.) or a summary of the patient's diagnosis and treatment.

A criminal finds out that a well-paid sports star is receiving treatment for mental health problems. He would like to gain illegal access to information in this system so that he can blackmail the star. By posing as a concerned relative and talking with the nurses in the mental health clinic, he discovers how to access the system and personal information about the nurses and their families. By checking name badges, he discovers the names of some of the people allowed access. He then attempts to log on to the system by using these names and systematically guessing possible passwords, such as the names of the nurses' children.

(a) Draw a use case diagram and show the processes where the criminal can be engaged to have access of the system. **(7)**

(b) Do you believe that the mental health clinic maintains proper IT security policy? Justify your answer. Design an IT security policy for the use of password for the clinic. **(4+10=14)**

(c) Design test cases for checking the proposed use of password policy stated in Question 1(b). **(10)**

(d) Find out assets, exposures and vulnerabilities for the Mentcare system stated above and show them in a table. **(6)**

(e) Write down the weaknesses of password-based authentication. Suggest appropriate 2FA and 3FA processes for the above case study for enhancing the access security. **(4+4=8)**

Contd P/2

CamScanner

CSE 213/CSE

2. (a) You are required to build a notification system named *StockMarket*. The system can generate *StockGrabber* objects. Each *StockGrabber* object sends notifications to all of its registered observers, only when a different price of a stock item is set. If the price is not changed, then setting the same value to any stock item does not generate any notification. When a *StockObserver* object gets registration under a *StockGrabber*, it gets an autogenerated *observerID* (starting from 1) and knows the prices of all stock items known to the associated *StockGrabber*. Assume for starting the system, when a *StockGrabber* is generated, it gets prices of three stocks items – *aplPrice* ($230 per unit), *googPrice* ($200 per unit) and *ibmPrice* ($180 per unit). A part of the program is given below.

```
public interface StockExchnage {
     public void register(StockObserver newObserver);
     public void unregister(StockObserver newObserver);
     public void notifyAllObserver(String company, double
                                   price);
}

public interface Observer {
    public void update(String company, double price);
}

public class StockMarket {
    public static void main(String [] args){
        StockGrabber securityCompany = new
                                   StockGrabber("Shamol");
        Observer observer1 = new
                    StockObserver(securityCompany, "Arif");
        Observer observer2 = new
                    StockObserver(securityCompany, "Kamal");
        securityCompany.setAplPrice(250.0);
        securityCompany.setGoogPrice(200.0);
    }
}
```

(i) Draw the appropriate UML diagram for the above descriptions using appropriate design pattern. **(6)**

(ii) Write down the codes for *StockGrabber* and *StockObserver* class so that no changes in the above code is required using appropriate Design Pattern. Ensure that the classes have minimum number of methods with simplified design and avoid any setters and getters which is unnecessary for the above code segment. **(8×2=16)**

(b) Read the following program descriptions carefully and draw the appropriate UML diagram using appropriate design pattern to represent them. **(8)**

Contd P/3

CamScanner

CSE 213/CSE

Contd ... Q. No. 2

The *HelpHandler* class defines the interface for handling help requests. It maintains a help topic (which is empty by default) and keeps a reference to its successor on the chain of help handlers. The key operation is *HandleHelp,* which subclasses override. *HasHelp* is a convenience operation for checking whether there is an associated help topic. All widgets (like *Dialog, Button)* are subclasses of the *Widget* abstract class. *Widget* is a subclass of *HelpHandler,* since all user interface elements can have help associated with them. *Button's* version of *HandleHelp* first tests to see if there is a help topic for buttons. If the developer hasn't defined one, then the request gets forwarded to the successor using the operation *HandleHelp* in *HelpHandler.* If there is a help topic, then the button displays it and the search ends. *Dialog* implements a similar scheme, except that its successor is not a widget but any help handler. This successor is an instance of *Application.* The Application is not a widget, so *Application* is subclassed directly from *HelpHandler.* When a help request propagates to this level, the application can supply information in general or it can offer a list of different help topics.

3. (a) Briefly describe test-driven development. Write down the benefits of test-driven development. **(6+6=12)**

(b) Generate test cases for white-box testing from the following code segment: **(10)**

```
FindMean (FILE ScoreFile){
      float SumOfScores = 0.0, Mean = 0.0, Score;
      int NumberOfScores = 0;

    Read(ScoreFile, Score);
    while (! EOF(ScoreFile) {
        if (Score  > 0.0 ) {
            SumOfScores = SumOfScores + Score;
            NumberOfScores++;
        }
        Read(ScoreFile, Score);
    }

/* Compute the mean and print the result */
    if (NumberOfScores > 0) {
        Mean = SumOfScores / NumberOfScores;
        printf(" The mean score is %f\n", Mean);
    } else
        printf ("No scores found in file\n");
    }
```

Contd P/4

CamScanner

CSE 213/CSE

Contd ... Q. No. 3

(c) Explain the basic principle of equivalence partition testing. Why is this testing used in a real-life software? **(8)**

4. (a) With appropriate diagram and examples show how security threats are arisen on instructions and data based on CIA Triad. **(7)**

(b) Explain encryption policy with appropriate diagram. Differentiate between symmetric encryption and asymmetric encryption. **(8)**

(c) An asset has an estimated value of $200,000 with an exposure factor of 20%. It is also estimated that frequency of attack on such an asset is once in every five-year time period. The actual cost of safeguards for the asset is $10,000 which reduces the exposure factor to 5%. Find out the residual risk that remains after taking the safeguard measures. Justify whether taking such a safeguard measure is worthy or not. **(7)**

(d) Explain buffer overflow attacks with appropriate example. **(8)**

SECTION – B

There are **FOUR** questions in this section. Answering Question 5 is mandatory.
Answer any **TWO** from question 6, 7 and 8.

5. (a) A software system is being developed for a pizza delivery service that allows customers to order pizzas with regional styles. The service operates in two countries, Italy and America, each with distinct preferences for pizza types. For instance, the crust type and toppings differ between regions. The system must ensure that each pizza is created with region-specific ingredients and should be easily extendable in the future to accommodate additional countries. Identify the most suitable design pattern to solve the given problem. Justify your choice and provide the corresponding code implementation. **(15)**

(b) Illustrate how a single class can serve as both an Adapter and a Facade with an example scenario. Design a class diagram to represent the scenario and analyze the functionality of the class in fulfilling the dual role. **(12)**

(c) Explain why achiving high testing coverage is important in software development. Mention the potential risks that might arise if coverage is too low. **(8)**

6. (a) A start-up company is developing a mobile health app that collects users' personal health data, such as heart rate, physical activity, and dietary habits. The app uses this data to provide critical health suggestions, such as warnings about potential heart risks or medicine recommendations for improving overall health. The project team consists of software and database engineers who have developed an algorithm by studying related articles to generate the suggestions. **(8)**

The app shares the health records as well as age, profession and address of the users with some healthcare institutions. However, prior to sharing, the app takes user consent to share anonymized health data.

Identify the ethical breaches present in this scenario (if any). Explain why these actions are unethical.

Contd P/5

CamScanner

CSE 213/CSE

Contd ... Q. No. 6

(b) "Software engineers shall act consistently with the public interest." - Elaborate the implication of this principle with relevant example. **(7)**

(c) A project manager reviews the progress of a project and discovers that for every dollar spent, he is getting 20% more of its worth work being done. The total project budget is $1,000. Encouraged by this efficiency, the manager is considering adding new features to the project. He has two options: Feature A would cost an additional $200, while Feature B would cost $100. Based on this information, determine which feature the manager can afford. **(20)**

Additionally, the manager finds that the actual cost incurred up to this point is $500. However, he did not review the planned value i.e., the value that was targeted to achieve at this stage, confidently assuming that the project is ahead of schedule. Verify whether his assumption is correct, providing the necessary calculations and reasoning.

7. (a) A global travel agency is planning to introduce a new AI-based feature aimed at enhancing customer experience. The feature will analyze customers' browsing and social media history to tailor recommendations. The agency is investing a significant portion of its budget into the development and rollout of this feature. It expects that it could substantially increase its customer base and boost sales. However, there are concerns about privacy, as many customers may be reluctant to share their browsing and social media data, which could lead to lower adoption rates. **(15)**

In addition to this, the agency is facing uncertainty due to the onset of a new epidemic that threatens their operations in several countries. If the epidemic becomes severe, government-imposed travel restrictions could force the agency to halt services in affected regions, resulting in a loss of revenue.

What types of risks can you identify in the scenario? For each risk, outline the appropriate risk management strategies that should be adopted and classify them as either proactive or reactive.

(b) In a highly reputed university, when exam results are published online, thousands of students simultaneously attempt to access the results portal. This surge in traffic often causes the portal to crash or become extremely slow. Despite increasing the number of servers, the issue persists. Analyze the possible underlying problem and propose an appropriate solution. **(10)**

(c) Describe the purpose and benefits of software profiling. Discuss the importance of monitoring resource utilization metrics and assessing the scalability of software during the development process. **(10)**

Contd P/6

CamScanner

CSE 213/CSE

8. (a) Describe a scenario where the Incremental Process Model is more suitable than the Agile Model for software development. Explain the factors that make the Incremental Process Model a better choice for the project, emphasizing the key characteristics of the project and the nature of its requirements. **(8)**

(b) Compare the roles of a manager and a leader. Provide a scenario where it is more beneficial to use a management approach rather than a leadership approach. **(7)**

(c) Define bad smells in code and code refactoring. Identify the bad smells present in the provided code below and revise it with appropriate refactoring techniques. **(20)**

```
class DiscountCalculator {
    public double calculateDiscount(String customerType, String
productType, double price, String customerName, String customerId) {

        System.out.println("Customer Name: " + customerName);
        System.out.println("Customer ID: " + customerId);

        if (customerType.equals("REGULAR") &&
productType.equals("ELECTRONICS")) {
            return price * 0.1; // 10% discount
        } else if (customerType.equals("REGULAR") &&
productType.equals("GROCERY")) {
            return price * 0.05; // 5% discount
        } else if (customerType.equals("PREMIUM") &&
productType.equals("ELECTRONICS")) {
            return price * 0.2; // 20% discount
        } else if (customerType.equals("PREMIUM") &&
productType.equals("GROCERY")) {
            return price * 0.1; // 10% discount
        } else {
            return 0;
        }
    }
}
```

P.T.O

Contd P/7

CSE 213/CSE

Contd ... Q. No. 8(c)

```
class OrderProcessor {
    public void processOrder(String customerType, String productType,
double price, String customerName, String customerId) {
        System.out.println("Processing Order for Customer: " +
customerName);
        System.out.println("Customer ID: " + customerId);

        DiscountCalculator discountCalculator = new
DiscountCalculator();
        double discount =
discountCalculator.calculateDiscount(customerType, productType,
price, customerName, customerId);

        if (customerType.equals("REGULAR")) {
            System.out.println("Regular customer processed");
        } else if (customerType.equals("PREMIUM")) {
            System.out.println("Premium customer processed");
        }

        if (productType.equals("ELECTRONICS")) {
            System.out.println("Electronics item processed");
        } else if (productType.equals("GROCERY")) {
            System.out.println("Grocery item processed");
        } else if (customerType.equals("PREMIUM")) {
            System.out.println("Other item processed");
        }
        System.out.println("Customer Name: " + customerName);
        System.out.println("Customer Type: " + customerType);
        System.out.println("Customer ID: " + customerID);

        System.out.println("Product Type: " + productType);
        System.out.println("Original Price: " + price);
        System.out.println("Discount: " + discount);
        System.out.println("Final Price: " + price - discount);

       if (discount == 0) {
           System.out.println("No discount applied");
       } else if (discount == price * 0.1) {
           System.out.println("10% discount applied");
       } else if (discount == price * 0.15) {
           System.out.println("15% discount applied");
       } else if (discount == price * 0.25) {
           System.out.println("25% discount applied");
       }
    }
}
```

— — — — —

CamScanner